

Nanoscale Imaging of Strain-Controlled Altermagnetic Domains in α -MnTe

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Abstract

Altermagnets combine compensated magnetic order with momentum-dependent spin splitting, offering a route to spintronic functionality without the stray fields of conventional ferromagnets. Mechanical strain provides a promising means of controlling their Néel order, yet the microscopic pathway by which strain reorganizes an altermagnetic texture remains unresolved. Here, we integrate a piezo-driven uniaxial strain cell with scanning nitrogen-vacancy magnetometry to image the magnetic domains of

bulk α -MnTe during in situ compression at room temperature. We find that compression reorganizes the magnetic texture through domain coalescence, increasing the size of the largest connected domain while reducing the domain-wall density. Upon unloading, however, the strain-formed domain network does not retrace the loading pathway. Instead, the large connected regions fragment into a new metastable configuration, producing pronounced hysteresis in the maximum domain size and stray-field distribution. These results identify domain connectivity and topology as key carriers of strain-induced magnetic memory. Our work reveals domain coalescence and hysteretic fragmentation as the microscopic pathway of strain control in α -MnTe and establishes a route toward strain-programmable altermagnetic textures and reconfigurable spintronic devices.

Introduction

Altermagnetism has recently emerged as a distinct form of magnetic order that combines the compensated real-space magnetization of antiferromagnets with momentum-dependent spin splitting more commonly associated with ferromagnets^{1–3}. The absence of a macroscopic magnetization suppresses stray fields and provides robustness against external magnetic perturbations, while the spin-split electronic structure enables spin-polarized transport and electrical readout. These properties make altermagnets promising candidates for dense, fast and robust spintronic technologies. Among the proposed altermagnetic materials, the g -wave altermagnet α -MnTe has become a prototypical platform because it hosts collinear compensated magnetic order above room temperature together with momentum-dependent spin splitting and an anomalous Hall response^{4,5}. Its hexagonal crystal structure and the coupling between the in-plane Néel vector and measurable magnetic and electronic responses provide an experimentally accessible setting in which to investigate how altermagnetic order can be detected and controlled.

A central requirement for allermagnetic spintronics is the deterministic manipulation of the Néel order. Strain engineering is particularly attractive because magnetoelastic coupling enables voltage-controlled modification of magnetic anisotropy without requiring large applied magnetic fields. Recent transport and neutron-scattering measurements have shown that uniaxial compression can modify the anomalous Hall response of α -MnTe and lift the degeneracy between crystallographically related magnetic variants^{6,7}. These observations have been interpreted as strain-induced detwinning, in which compression preferentially populates one of the three in-plane Néel axes and drives the system toward a single-domain state. By contrast, recent magneto-optical measurements suggest that the twofold magnetoelastic anisotropy can overcome the intrinsic sixfold magnetocrystalline anisotropy and continuously rotate the local Néel vector⁸. These pictures emphasize different aspects of the strain response: one describes the resulting redistribution of magnetic-domain populations, whereas the other describes the local angular evolution of the order parameter. However, both interpretations have largely been inferred from spatially averaged measurements rather than direct imaging of the nanoscale magnetic domains, leaving unresolved how strain-driven Néel-vector reorientation is realized through the motion, merging and reorganization of individual domains. This distinction is particularly important in α -MnTe because the weak out-of-plane moment is related to the in-plane Néel-vector angle through $M_z \propto \sin(3\phi_L)$, such that each sign of the out-of-plane moment represents three possible 120°-separated orientations of $\mathbf{L}$ ^{5,9,10}. Direct real-space imaging under continuously tunable strain is therefore required to determine the domain dynamics that connect local Néel-vector reorientation to the formation of a strain-selected magnetic state.

Here, to address this debate, we integrate a piezo-driven uniaxial strain cell with scanning nitrogen-vacancy (NV) magnetometry, enabling the same microscopic region to be imaged while compressive strain is varied in situ. Using this platform, we follow the room-temperature magnetic-domain texture of bulk α -MnTe through a complete compression–release cycle. Increasing compression promotes domain reorientation and coalescence, in-

creasing the maximal domain size while reducing the domain-wall density. This evolution provides a direct microscopic pathway linking strain-induced Néel-vector reorientation to the progression toward a strain-selected single-domain state. During unloading, however, the domain network follows a different trajectory and fragments into a metastable configuration, producing pronounced hysteresis in the maximum domain size and local stray-field distribution despite comparatively weak hysteresis in the domain wall density. Our observations therefore connect the recently proposed pictures of continuous Néel-vector rotation and strain-induced detwinning by showing that strain control is realized through domain-boundary motion, coalescence, pinning and metastability. These results establish scanning NV magnetometry under in situ strain as a platform for resolving and engineering altermagnetic-domain dynamics at the nanoscale.

Experimental results

α -MnTe crystallizes in the hexagonal NiAs structure and hosts a collinear compensated antiferromagnetic order, in which Mn^{2+} moments are aligned ferromagnetically within each basal-plane layer and antiferromagnetically between adjacent layers along the c axis^{5,9,10}. Although the antiferromagnetic spin sublattices are nearly compensated, symmetry permits a weak net magnetic moment along the out-of-plane direction. The microscopic origin of this weak out-of-plane (OOP) moment remains an active subject of discussion, particularly regarding the relative roles of spin canting, orbital magnetization, surface symmetry breaking, and higher-order spin-orbit-coupled interactions¹¹. Within recent symmetry-based descriptions, the weak OOP moment is coupled to the in-plane Néel vector through a higher-order spin-orbit-coupled invariant, rather than through a conventional Dzyaloshinskii–Moriya-type spin-canting mechanism that would generate an effective bilinear coupling between $\mathbf{L}$ and $\mathbf{M}$ ¹². As a result, the sign of the weak OOP moment is locked to the in-plane orientation of $\mathbf{L}$, but the measured sign of M_z does not uniquely specify the full in-plane Néel-vector

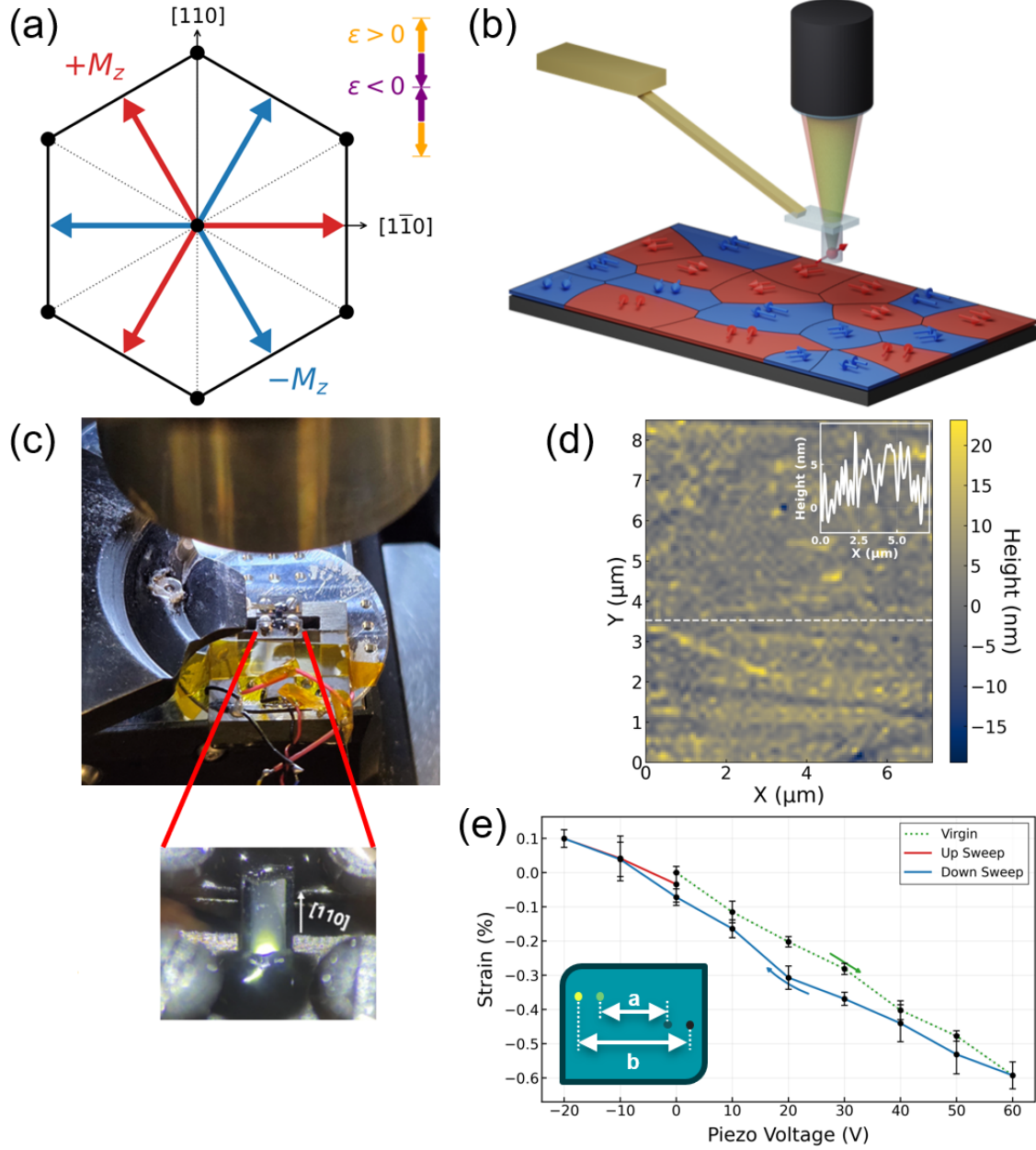


Figure 1: (a) Schematic of MnTe crystal structure. Mn atoms are shown as black circles and Te atoms are omitted for clarity. Red/blue arrows correspond to the 6 Néel vector easy axes, each associated with one of two OOP magnetic moments $\pm M_z$. Purple/orange arrows highlight compressive/tensile strain directions respectively along [110] axis. In this work we mainly studied the effect of compressive strain. (b) Schematic of scanning NV measurement. A diamond-tipped probe containing a single NV is scanned over a sample consisting of $\pm M_z$ regions (red/blue) associated with various Neel vector orientations represented by the in-plane arrows. Excitation laser light is applied and NV PL is collected through a microscope objective above to optically readout the stray field strength. (c) Optical image of MnTe strain cell applying strain along [110] axis. (Inset) Zoomed-in image of the same MnTe crystal with [110] direction labeled. (d) AFM image of ROI used in all subsequent figures. (Inset) Linecut of AFM scan across horizontal dashed white line. (e) Calibration curve of voltage to strain determined by large area AFM scans [see Supplementary Note 1]. (Inset) Strain determined as the relative change of distance between particles on surface $\varepsilon = \frac{b-a}{a} \times 100\%$.

direction.

Each sign of the OOP moment corresponds to three possible Néel vector orientations as depicted schematically in Fig. 1a. The six easy axes point along the Mn ion next-nearest-neighbor (NNN) directions, with adjacent directions corresponding to an opposite OOP ferromagnetic moment. Therefore, a given stray field actually sensed by the NV center does not itself specify a particular Néel vector direction. Conversely, the selection of a single in-plane Néel-vector axis does not necessarily produce a single sign of M_z . Each in-plane axis contains two time-reversal-related states, $\mathbf{L}$ and $-\mathbf{L}$, which carry opposite weak OOP moments. Thus, the single-domain state identified by orientation-sensitive measurements refers to the selection of one in-plane Néel axis, whereas positive- and negative- M_z regions can remain within that selected axis pair. This degeneracy is central to the interpretation of our results: e.g., our technique is not sensitive to domain walls between Néel vector orientations that share the same sign of M_z . While this fact bounds certain types of inferences, others can still be made through careful scrutiny of the OOP moment distributions.

Figure 1b shows the experimental setup: a single NV center at the tip of a scanning probe senses the projection of the stray field $B_{\parallel}(x, y) = \mathbf{B}(x, y) \cdot \hat{\mathbf{n}}_{\text{NV}}$ on the NV axis $\hat{\mathbf{n}}_{\text{NV}}$ as a function of position above the sample and at a height of ~ 40 nm. The OOP magnetic field $B_z(x, y)$ is calculated from $B_{\parallel}$ using a Fourier-space reconstruction that enforces the source-free condition $\nabla \cdot \mathbf{B} = 0$ above the sample and the known NV spin axis. The effective OOP magnetization $M_z(x, y)$ can then be estimated via a gradient-descent algorithm, using a loss function comprised of a term given by the difference between the original $\mathbf{B}(x, y)$ and the forward calculated $\mathbf{B}_{\text{fw}}(x, y)$ from the guessed magnetization, plus two terms related to the magnitude and change in the magnetization vector (see Supplementary Note 2)¹³. However since the $B_z(x, y)$ and $M_z(x, y)$ maps are qualitatively similar, and because neither quantity uniquely determines the underlying in-plane Néel-vector orientation, most data and inferences drawn in this work rely on the OOP field $B_z(x, y)$.

To manipulate the magnetic free energy landscape, the sample was mounted within a

strain cell (Figure 1c) capable of applying precise compressive strain along the $[110]$ crystallographic direction. The sample surface was polished, and an atomic force microscopy (AFM) scan was taken at the location that magnetometry was performed, showing a minimal height variation of $h_{\text{rms}} = 3.7 \text{ nm}$ (Fig. 1d). AFM scans over a much larger area (see Supplementary Note 1) were used to track the positions of particles at different voltages applied to the strain cell, allowing the sample strain to be calibrated (Figure 1e).

Figure 2 presents the spatial evolution of the magnetic texture across a complete compressive loading–unloading cycle. The NV magnetometry maps in Figure 2a reveal a complex multidomain landscape established following the initial out-of-plane field-cooling procedure. As compressive strain is swept along the $[110]$ axis and subsequently released, the local OOP stray-field texture undergoes substantial reorganization and does not return to its initial configuration. The mean value of B_z shifts toward more positive values under increasing compression and moves back toward more negative values as the compression is released, without fully recovering its initial value (Figure 2b inset). Likewise, interpolated linecuts in Figure 2c show the expansion of positive- B_z regions at large compressive strain and their recession or fragmentation on the release branch.

To quantify this reorganization, Figure 2d tracks the maximum positive-domain area and the total positive area contained in domains larger than $5 \mu\text{m}^2$. Both quantities increase during the loading branch, demonstrating that compression promotes the formation of large connected positive- B_z regions. This behavior indicates that the positive projected OOP-moment sector does not grow primarily through the nucleation of many isolated small domains. Instead, compression drives boundary-mediated coalescence, in which neighboring regions with compatible projected M_z sign merge into larger connected domains. This coalescence picture is further supported by the decrease in the visible domain-wall density under compression (Figure 3a), which indicates that intervening boundaries are eliminated as neighboring regions merge into larger same-sign domains. The simultaneous increase of the largest connected positive domain and the total large-domain area shows that the upper tail

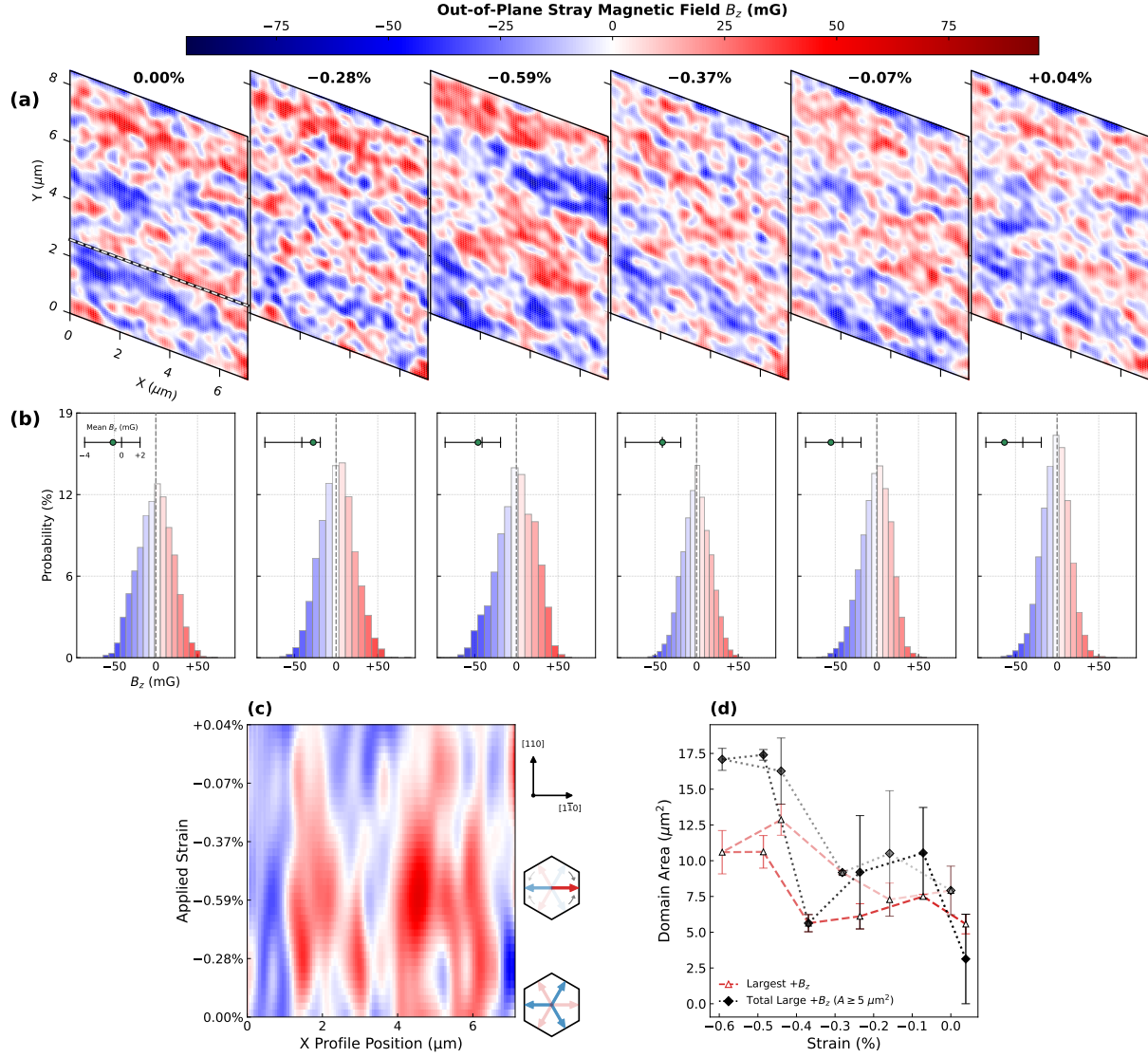


Figure 2: (a) B_z component extracted from NV magnetometry scans during a compression–release cycle, from 0% strain to -0.6% (compressive), and back to 0.04% . Dashed white line in left-most scan represents location of linecuts presented in (c). (b) Associated B_z histograms of corresponding to scans in (a). Approximately 10,000 data points were used to form each histogram. (Inset) Mean value of B_z . (c) Interpolated linecut comparison of B_z values [white/black dashed line in (a)] as strain is varied. Inset: schematic possible reorientation pathway, with relative population indicated qualitatively by arrow opacity. OOP field cooling initially biases the population toward the three $-M_z$ variants, while compression selects the strain-favored in-plane axis pair. If local regions reorient toward the nearest strain-favored direction, this can produce an unequal population of the two time-reversal-related states and increase the observed positive- M_z sector. (d) Maximum positive-domain area and total large-domain area, the latter defined as the total positive area contained in domains larger than $5\mu\text{m}^2$. Faded lines correspond to earlier measurements, progressing to solid lines for later measurements.

of the positive-domain size distribution grows under compression. The texture nevertheless remains multidomain over the strain range studied here, indicating that coalescence is limited by pinning and disorder rather than proceeding to complete monodomain formation. The release branch reveals that this coalescence is not simply reversed. When the compression is reduced from its maximum value, the largest connected positive domain decreases rapidly and became smaller than in the initial nominally unstrained state. The total large-domain area follows a similar hysteretic trend. Thus, unloading appears to fragment, or depercolate, the strain-formed positive-domain network into a new metastable configuration rather than restoring the original domain pattern.

A simple geometric picture (inset of Fig. 2c) illustrates how an initially field-cooled OOP imbalance can nevertheless produce the observed increase of the positive- M_z sector under compression. While the initial OOP field-cooling procedure did not fully saturate the sample, an observable bias toward $-M_z$ can be seen in Fig. 2. Compressive strain lowers the energy of the Néel-vector axis pair associated with the strain-favored direction, in this case along $\pm[1\bar{1}0]$. Assuming that the three $-M_z$ Néel-vector variants are approximately equally populated, if local regions reorient toward the nearest member of this strain-favored axis pair, the two final time-reversal-related directions need not acquire equal population (inset of Fig. 2c). Specifically, one of the two selected directions can inherit population from two initially majority $-M_z$ variants and one initially minority $+M_z$ variant, while the opposite selected direction inherits population from one majority variant and two minority variants. Thus, an initially OOP-biased but in-plane-degenerate texture can be converted by compressive strain into an unequal population of the two time-reversal-related states within the selected in-plane axis pair. This provides a possible microscopic pathway by which strain-induced axis selection, together with the memory of the field-cooled OOP imbalance, produces the observed growth of the positive- B_z sector (see Supplementary Note 3 for more details).

Taken together, these observations provide a real-space pathway that unifies the detwinning and continuous-rotation descriptions. In the continuous-rotation picture, strain changes

the preferred local angle of the Néel vector. In the detwinning picture, strain changes the relative populations of the crystallographically related Néel-vector variants. Our measurements show how these two descriptions can be connected microscopically: local Néel-vector reorientation is realized through domain-boundary motion, i.e., coalescence, and fragmentation of the projected M_z texture. During loading, compression promotes the merger of kinetically susceptible regions into larger connected domains, increasing both the maximum domain area and the total large-domain area. During unloading, the strain-formed connected network does not retrace the loading pathway, but instead fragments or depercolates into a new metastable configuration. Thus, the strain response is not adequately described as a spatially uniform rotation of an otherwise fixed texture, nor as an instantaneous homogeneous collapse into a complete monodomain state. Rather, strain-induced Néel-vector reorientation proceeds through a heterogeneous domain network whose connectivity and memory are controlled by pinning and metastability.

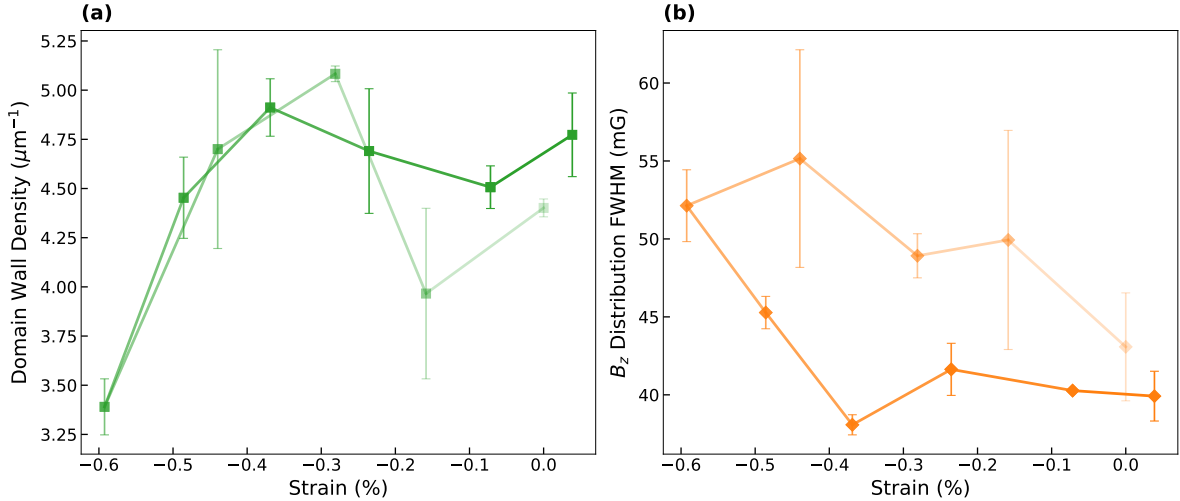


Figure 3: (a) Average domain wall density and (b) FWHM of B_z distributions (see Fig. 2b), as functions of compressive strain along [110]. Faded lines correspond to the first earlier measurements, with more solid lines representing later measurements.

The same coalescence–fragmentation picture provides a natural link between the large-domain statistics in Figure 2d and the field-distribution metrics analyzed in Figure 3. Because the magnetic stray field is measured at a finite NV–sample standoff, small or highly

intermixed regions of opposite OOP moment partially cancel and produce reconstructed B_z values closer to zero. During loading, compression promotes the coalescence of smaller same-sign regions into larger connected domains. This reduces cancellation between closely intermixed positive- and negative- B_z regions and increases the fraction of pixels with larger absolute reconstructed field values. Probability is therefore transferred from the central part of the histogram toward the sides, producing the broader B_z distribution observed on the increasing-compression branch in Figure 3b. During unloading, the strain-formed connected regions fragment or lose connectivity as the magnetoelastic driving force is reduced. The resulting finer mixture of positive- and negative- B_z regions increases local cancellation and narrows the B_z distribution. Thus, the matching hysteresis of the largest-domain size and the B_z -distribution width supports a common origin in loading-induced coalescence and unloading-induced breakup of large connected domains.

To further quantify the strain-driven domain evolution, we extract statistical metrics from the reconstructed B_z maps, as summarized in Figure 3. The domain-wall density plotted in Figure 3a was extracted from the B_z domain masks and provides a measure of the visible boundary length per unit area. This quantity exhibits a weakly non-monotonic response at low-to-intermediate compression, followed by a clearer decrease at larger compression. One possible interpretation is that, before net coalescence dominates, the evolving magnetoelastic anisotropy first mobilizes and reshapes the existing domain network, allowing boundaries to move, reconnect, and locally increase the total visible wall length. At larger compression, where the strain-induced anisotropy more strongly selects favored domain orientations, coalescence and boundary elimination become dominant, producing the observed reduction in domain-wall density. The weak hysteresis of this metric indicates that the total visible boundary length is governed mainly by the instantaneous applied strain, rather than by the long-lived memory encoded in the domain-network connectivity. Thus, the domain-wall-density trend is consistent with a crossover from strain-induced wall reconfiguration at low-to-intermediate compression to net boundary elimination at larger compression, rather

than with uniform coarsening throughout the full strain range.

The FWHM of the B_z distribution captures the global spread of local OOP stray fields within the scan area (Figure 3b). During loading, the coalescence of smaller domains produces larger connected regions of the same B_z sign. Finely intermixed positive- and negative- B_z regions partially cancel at the NV measurement height and consequently contribute a larger number of values near the center of the histogram. When smaller domains merge into larger same-sign regions, this cancellation is reduced and more pixels acquire larger positive or negative field amplitudes. Probability is therefore transferred from the central bins toward the sides of the distribution, producing the observed increase in the approximately Gaussian B_z FWHM under compression. During unloading, the strain-formed connected regions fragment as the magnetoelastic driving force is reduced. The resulting finer mixture of positive- and negative- B_z regions increases local field cancellation and transfers probability back toward the center of the histogram, narrowing the B_z distribution. Because the unloading process does not reconstruct the original domain configuration, the return-branch FWHM can fall below its initial value. The matching hysteresis of the largest-domain size and B_z -distribution width supports a common origin in the loading-induced formation and unloading-induced breakup of large connected domains.

Taken together, the metrics in Figure 3 separate the strain response of the total visible boundary length from the response of the stray-field amplitude distribution. The domain-wall density is comparatively weakly hysteretic, indicating that the total amount of visible positive/negative boundary per unit area is not the primary memory variable during the strain cycle. By contrast, the largest-domain metrics in Figure 2d and the B_z -distribution width in Figure 3b are strongly hysteretic.

This contrast shows that similar total visible boundary lengths can coexist with substantially different domain configurations. The strain memory is carried primarily by the connectivity and organization of the domain network, rather than simply by the total amount of domain wall. The strain cycle is therefore comparatively reversible in total visible bound-

ary length, but strongly hysteretic in the connectivity and organization of the projected OOP-moment texture and in the resulting distribution of stray-field amplitudes.

Discussion and Outlook

We have developed scanning NV magnetometry under in situ uniaxial strain and used it to directly visualize the room-temperature domain dynamics of bulk α -MnTe. Our measurements show that compressive strain along the $[110]$ direction drives domain reorientation and coalescence in bulk α -MnTe, producing larger connected domains and reducing the domain-wall density. During unloading, the domain pattern follows a different pathway and becomes trapped in a new metastable configuration, revealing pronounced magnetic memory.

Overall, our work provides direct microscopic insight into the ongoing debate over whether strain rotates the Néel vector or drives α -MnTe toward a single-domain state. Our results connect these two pictures by showing that local Néel-vector reorientation and domain coalescence provide a real-space mechanism toward a strain-selected magnetic state, whereas pinning and metastability govern the history-dependent reconfiguration during unloading. These observations therefore help resolve the apparent disagreement by establishing that Néel-vector rotation and single-domain formation are not necessarily competing mechanisms, but can represent different stages of the same strain-driven domain evolution.

The observed real-space hysteresis also provides a microscopic basis for the persistent strain dependence of the anomalous Hall effect in α -MnTe^{6,7}. Because the anomalous Hall conductivity depends on the angular distribution of the in-plane Néel order, spatial averaging over differently oriented domains can reduce, enhance or reverse the macroscopic response^{5,6}. Compression-driven selection, continuous local rotation or a combination of the two can therefore modify the anomalous Hall signal by changing both the populations and orientations of the magnetic variants. Our observation that one projected OOP-moment sector expands and becomes less fragmented under increasing compression provides a direct

real-space mechanism for such a redistribution. More importantly, the incomplete recovery during unloading suggests that the transport response may encode not only the instantaneous applied strain but also the metastable domain configuration selected by the preceding strain trajectory. The anomalous Hall signal could therefore retain memory after the externally imposed anisotropy has been reduced. At the same time, the NV measurement probes the weak OOP moment rather than the Berry curvature or anomalous Hall conductivity directly, and the many-to-one relation between M_z and ϕ_L prevents a one-to-one assignment between an individual local magnetic feature and the macroscopic Hall response. Simultaneous scanning magnetometry and transport measurements on the same strained device would directly determine how individual domain rearrangements contribute to the Hall signal and its hysteresis.

Future measurements combining scanning NV magnetometry with orientation-sensitive imaging could directly relate local domain rearrangements to the Néel-vector angle and anomalous Hall response. Time-resolved imaging and repeated strain cycles would further quantify domain-wall depinning, relaxation and training. Beyond α -MnTe, this capability provides a platform for mechanically programming compensated magnetic textures. The observed history dependence could enable strain-written nonvolatile or multilevel memory, reconfigurable altermagnetic states and programmable spintronic networks controlled electrically through piezoelectric actuation. More broadly, the ability to image compensated magnetic textures while continuously tuning their mechanical energy landscape establishes a route for understanding and ultimately engineering the nonequilibrium domain processes that govern strain-based altermagnetic memory, reconfigurable spintronics and mechanically programmable quantum materials.

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